

```
data <- ChickWeight

# Convert Diet to factor
data$Diet <- factor(data$Diet)

# Multiple regression
model <- lm(
  weight ~ Time + Diet,
  data = data
)

cat("Regression Model:\n")
print(summary(model))

# User input
time <- as.numeric(
  readline("Enter Time: ")
)

diet <- readline(
  "Enter Diet (1/2/3/4): "
)

new_data <- data.frame(
  Time = time,
  Diet = factor(
    diet,
    levels = levels(data$Diet)
  )
)

# Prediction
prediction <- predict(
```

```

model,

newdata = new_data

)

cat("\nPredicted Weight =", prediction)

# Model error

predicted_all <- predict(model, data)

error <- mean(

  (data$weight - predicted_all)^2

)

cat("\nMean Squared Error =", error)

```

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for predicting weight based on time and diet, calculating the Mean Squared Error (MSE).
- Environment:** Shows the objects created in the global environment:
 - `data`: 578 obs. of 4 variables
 - `model`: List of 13
 - `file_name`: "2"
- Console:** Displays the output of the R code execution, including the regression model summary:


```

> source("C:/Users/2021e/Desktop/EX 1.R")
Regression Model:

Call:
lm(formula = weight ~ Time + Diet, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-136.851  -17.151   -2.595   15.033  141.816

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.9244    3.3607   3.251  0.00122 **
Time          8.7505    0.2218  39.451 < 2e-16 ***
Diet2        16.1661    4.0858   3.957  8.56e-05 ***
Diet3        36.4994    4.0858   8.933 < 2e-16 ***
Diet4        30.2335    4.1075   7.361  6.39e-13 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 35.99 on 573 degrees of freedom
Multiple R-squared:  0.7453,    Adjusted R-squared:  0.7435
F-statistic: 419.2 on 4 and 573 DF,  p-value: < 2.2e-16

```